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**Moses Nandwa** · Follow

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What is Splunk?

The word Splunk comes from the word Spelunking which means exploring underground caves. In our case Splunk means exploring machine data thus making it accessible, usable and valuable to everyone in realtime. Machine data contains critical insights which when analysed can be used to answer different questions and give insights.

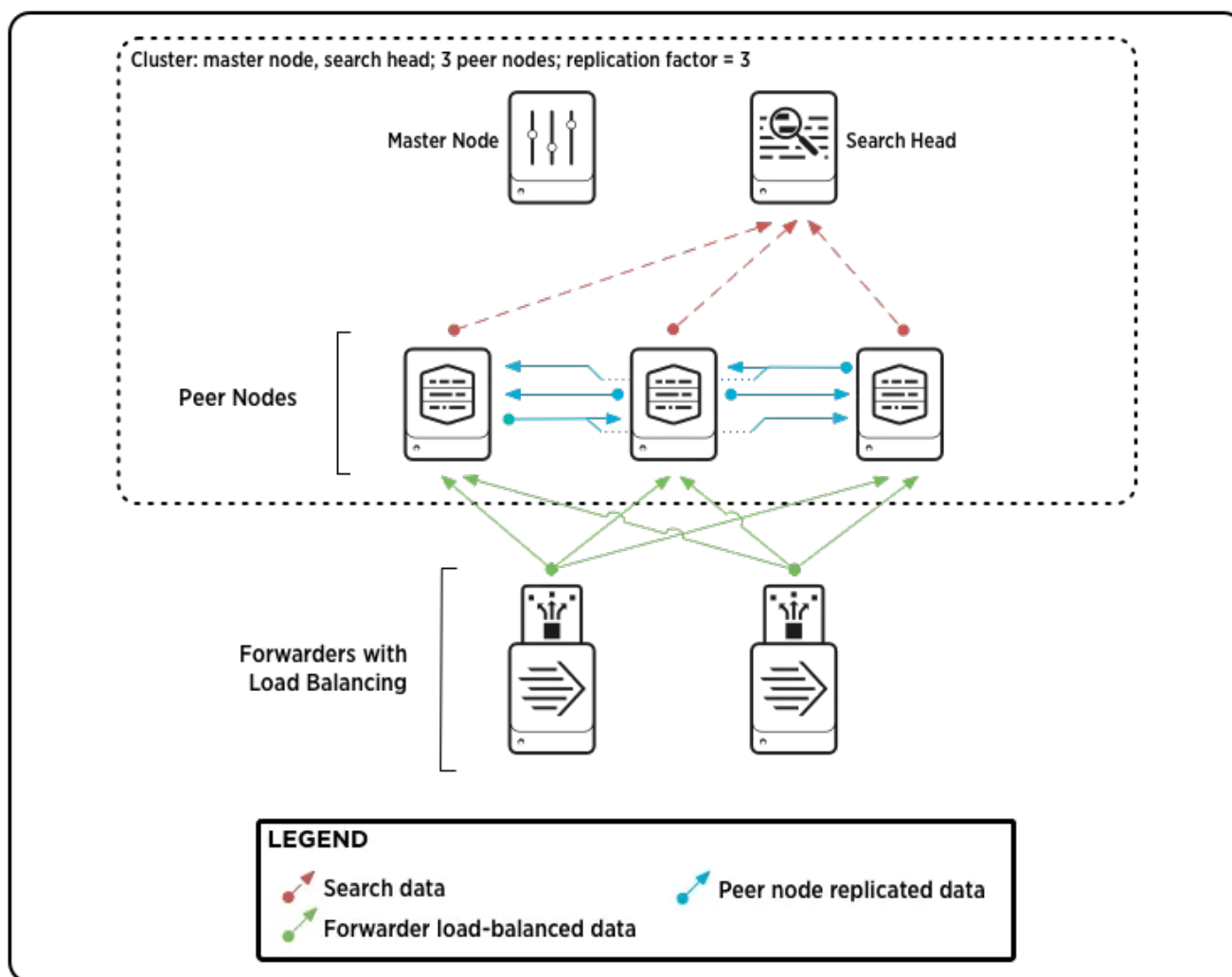
Splunk indexer cluster

An indexer cluster is a group of Splunk Enterprise instances, or nodes , that, working in concert, provide a redundant indexing and searching capability.

Typically, a cluster consists of:

1. **Master Node** — Responsible for managing the cluster
2. **Search heads** — They are used to conduct searches across the peer nodes
3. **Peer Nodes** — They maintain multiple copies of data. They receive data from

forwarders and applications.



The structure of a Splunk indexer cluster

In this tutorial we shall set up a Splunk cluster, push data from a Docker container into it and conduct a search to show the container's Docker logs.

Splunk Installation

Splunk can be installed directly on a host by using downloaded Splunk RPMs , DEB, or tar files or using Docker. In our case we shall use Docker. In this case we will require 4 Virtual machines with Docker installed on them. The machines shall be used for the following purpose:

1. Master
2. Peer node

3. Search head

4. App server

Run the following command in the Master, peer node, search head to start Splunk on all the VMs.

```
docker run -d -p 8080:8080 -p 8000:8000 -p 9997:9997 -p 8088:8088 -p 8089:8089 -e 'SPLUNK_START_ARGS= -- accept-license' -e 'SPLUNK_PASSWORD=Kz@asw57' splunk/splunk
```

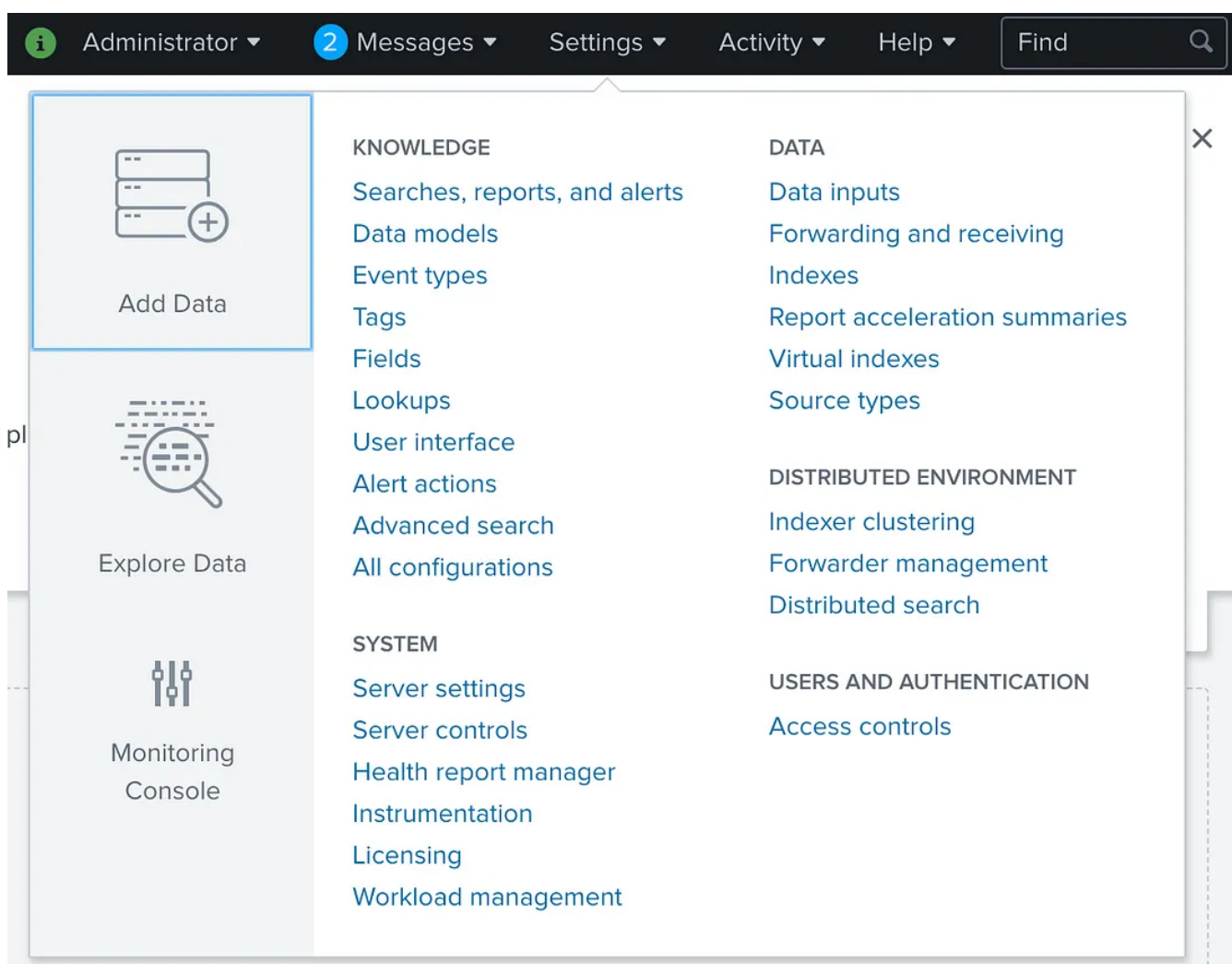
The command above starts Splunk, accepting the licence , setting the admin login password to *Kz@asw57* and opening up several ports that Splunk uses. Here are the uses of the Splunk ports:

1. **9997** — for forwarders to the Splunk indexer.
2. **8000** — Access to the web interface
3. **8089** — Connects to the Data Collection Node (DCN) on the default Splunk management port, TCP 8089.
4. **8088** — Receives data from Http event collectors that will be used to collect data from Docker containers
5. **8080** -Replication of data in peer nodes

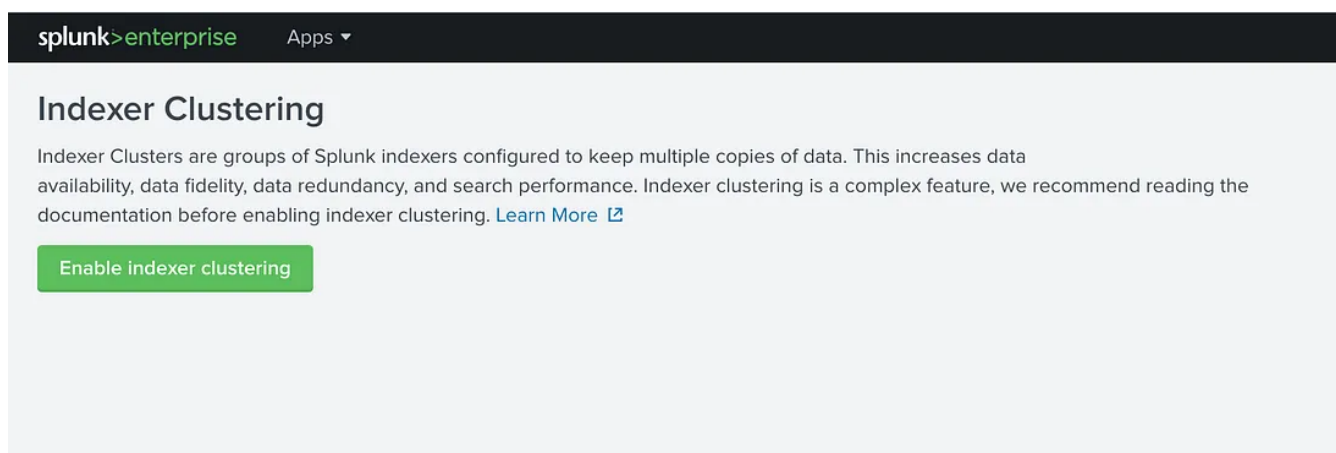
Cluster Configuration

Configuring the master node

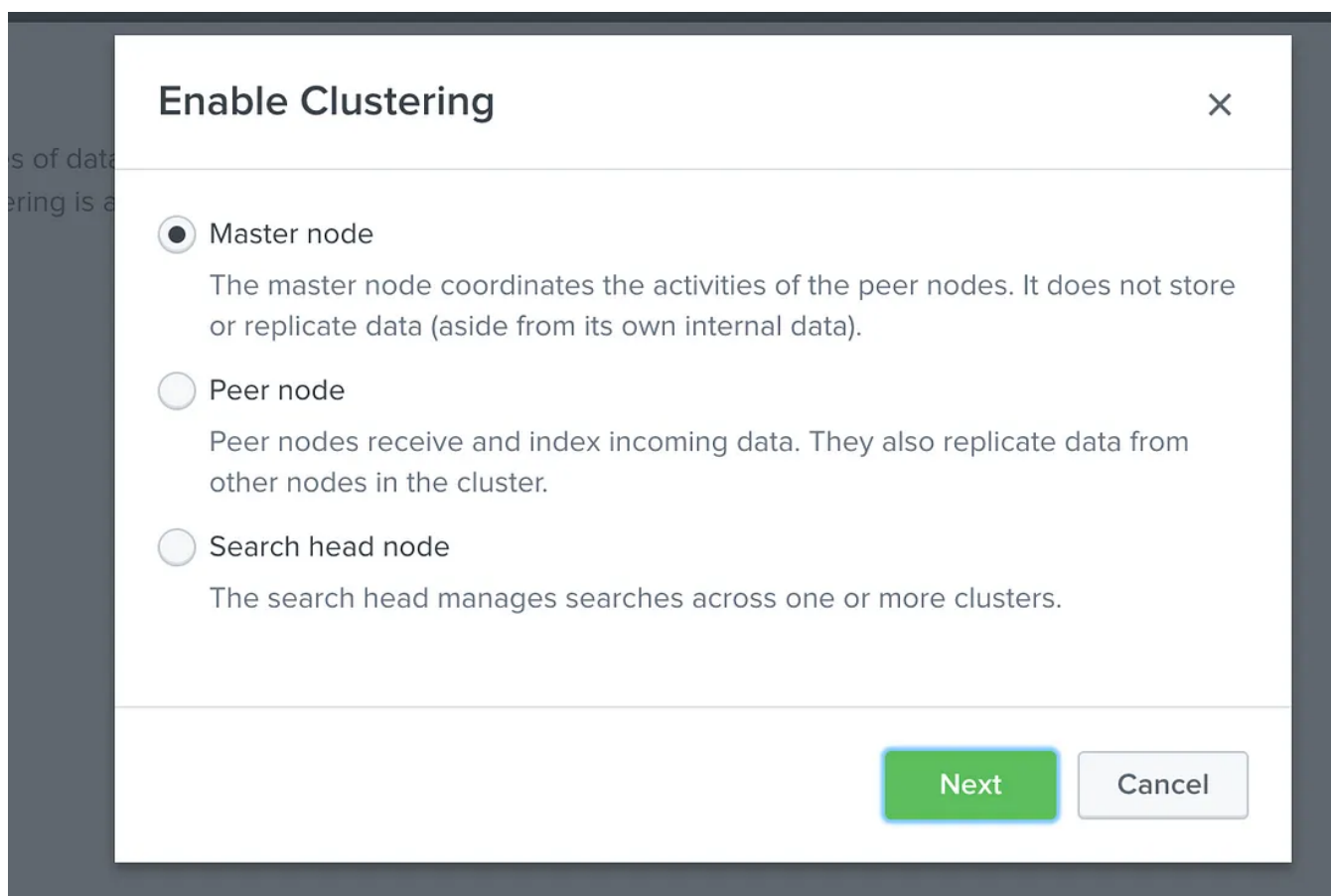
Log into the master node from the browser using the *admin* as the username and the password that was set during container creation. Once inside head to the top-right corner of the window and click *Settings > Indexer clustering*.



In the new page click *Enable Indexer clustering*



In the modal box that appears, select *master node* and click *next*.



For now we shall use a replica factor of one and a search factor one for learning purposes. However it is wise to have more incase one peer node goes down. In this window set a security key that the search head and the peer nodes will use to connect to the cluster.

Master Node Configuration

×

Replication Factor

The number of copies of raw data that you want the cluster to maintain. A higher replication factor protects against loss of data if peer nodes fail.

Search Factor

The number of searchable copies of data the cluster maintains. A higher search factor speeds up the time to recover lost data at the cost of disk space. Must be less than or equal to Replication Factor.

Security Key

This key authenticates communication between the master and the peers and search heads.

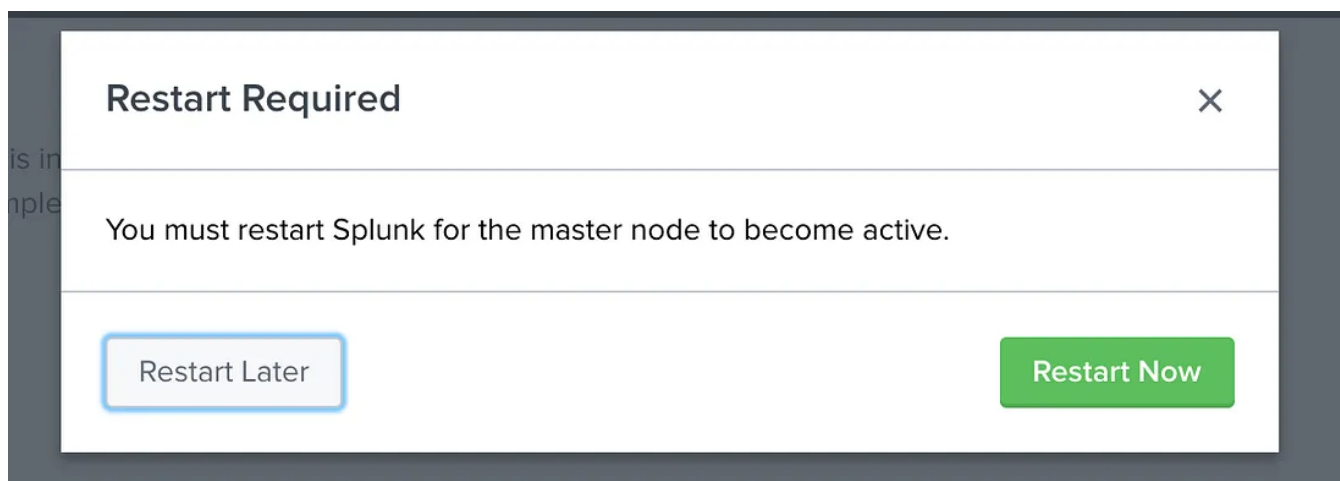
Cluster Label

Name your cluster using this field. This label is also used to identify this cluster in the Monitoring Console.

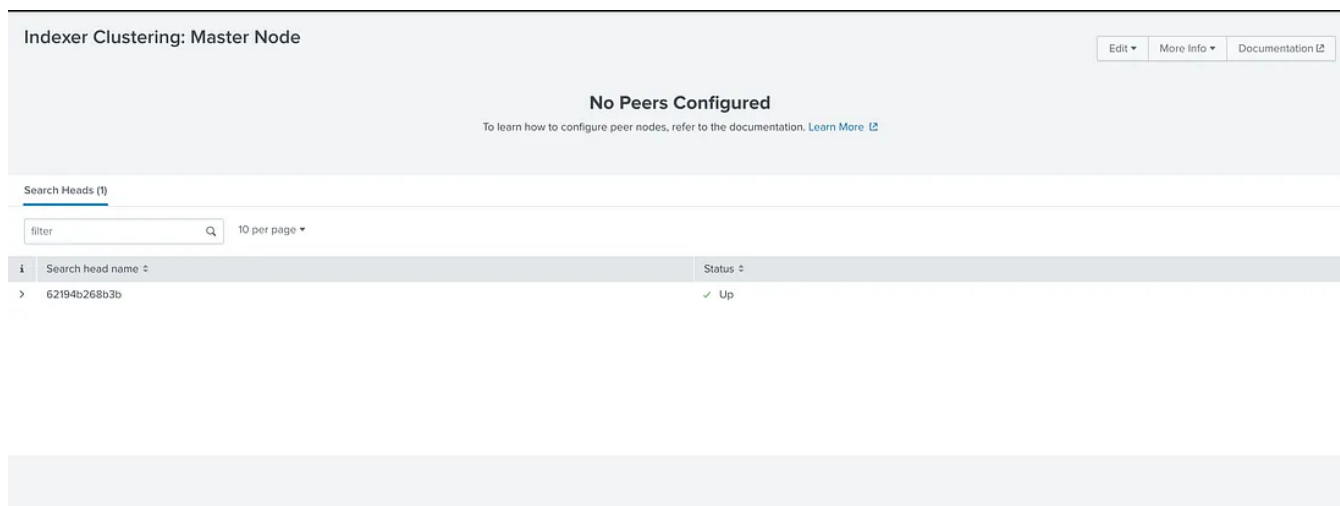
Cancel

Save changes

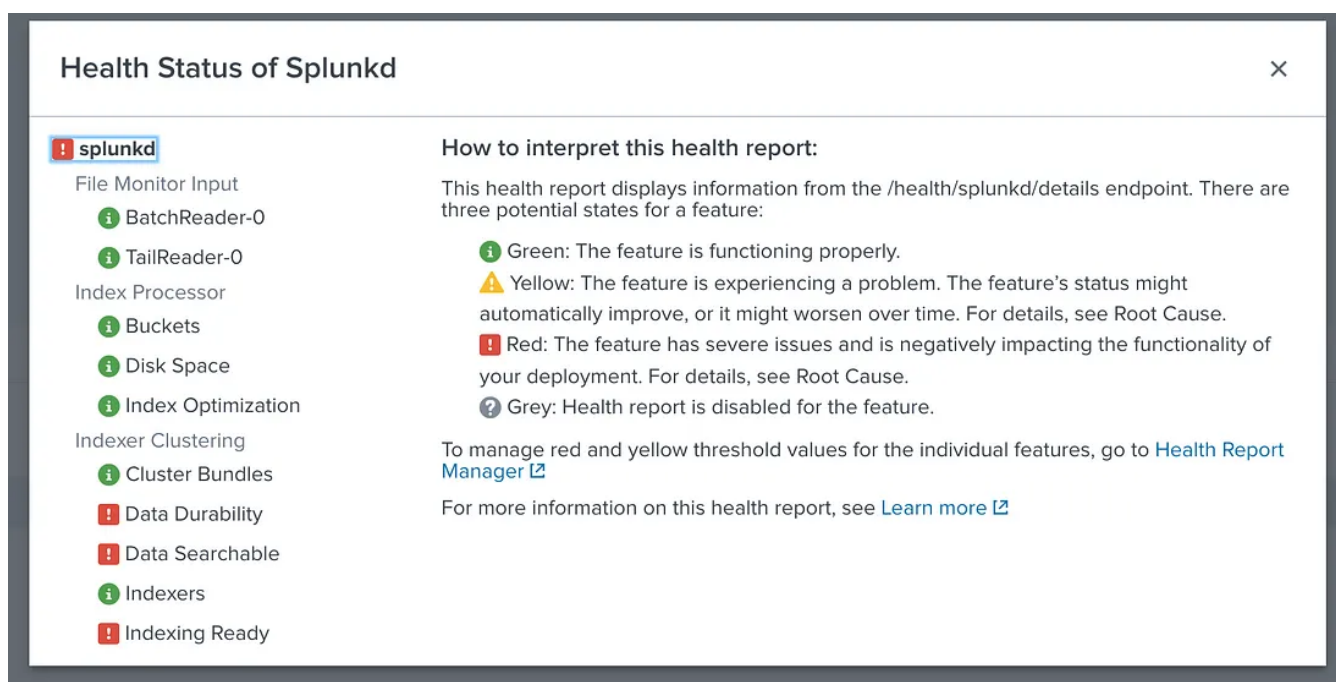
Restart for the settings to come to effect.



The result is this window.

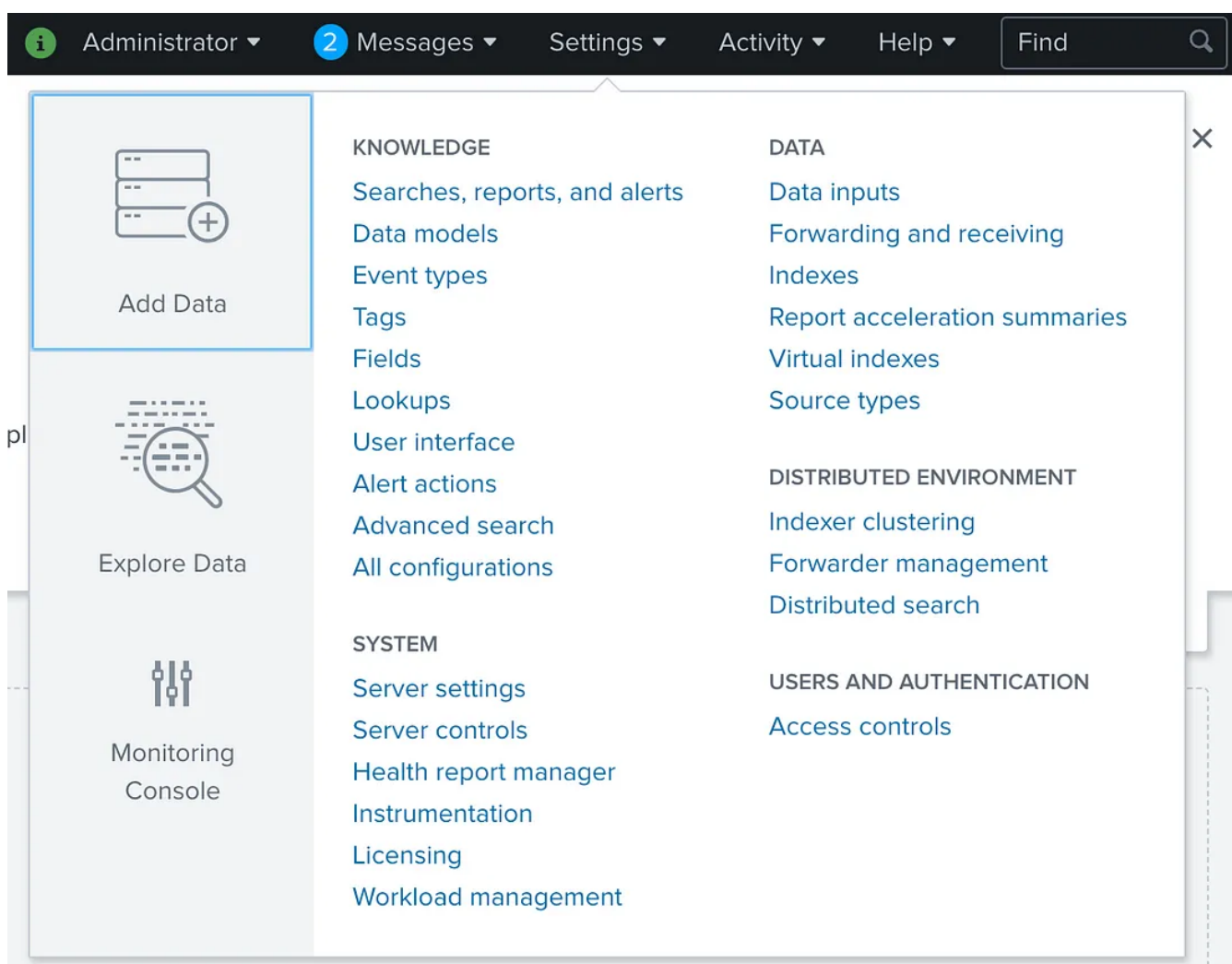


There will also be some errors on the Splunk health status arising from the fact that we have not configured the indexers. Do not worry about them for now as they will disappear when we finish configuring the cluster.

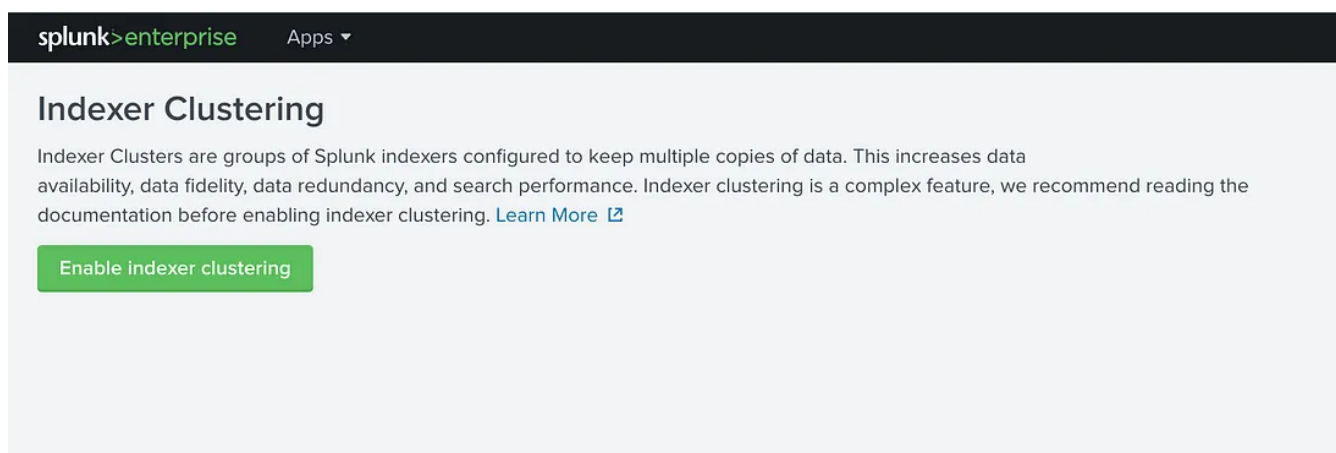


Configuring the Search head

Log into the master node from the browser using the ‘admin’ as the username and the password that was set during container creation. Once inside head to the top-right corner of the window and click *Settings > Indexer clustering*.



In the new page click *Enable Indexer clustering*



On the pop up modal that appears, select *search head node* and click *next*

Enable Clustering

Master node

The master node coordinates the activities of the peer nodes. It does not store or replicate data (aside from its own internal data).

Peer node

Peer nodes receive and index incoming data. They also replicate data from other nodes in the cluster.

Search head node

The search head manages searches across one or more clusters.

Next

Cancel

Enter the *Master URI* and the port. By default, Splunk use port *8089* to connect to the master

Search head node configuration

Master URI

E.g. https://10.152.31.202:8089 This can be found in the Master Node dashboard.

Security key

This key authenticates communication between the master and search head.

Back

Enable search head node

Restart for the settings to come to effect.

Restart Required

You must restart Splunk for the master node to become active.

Restart Later

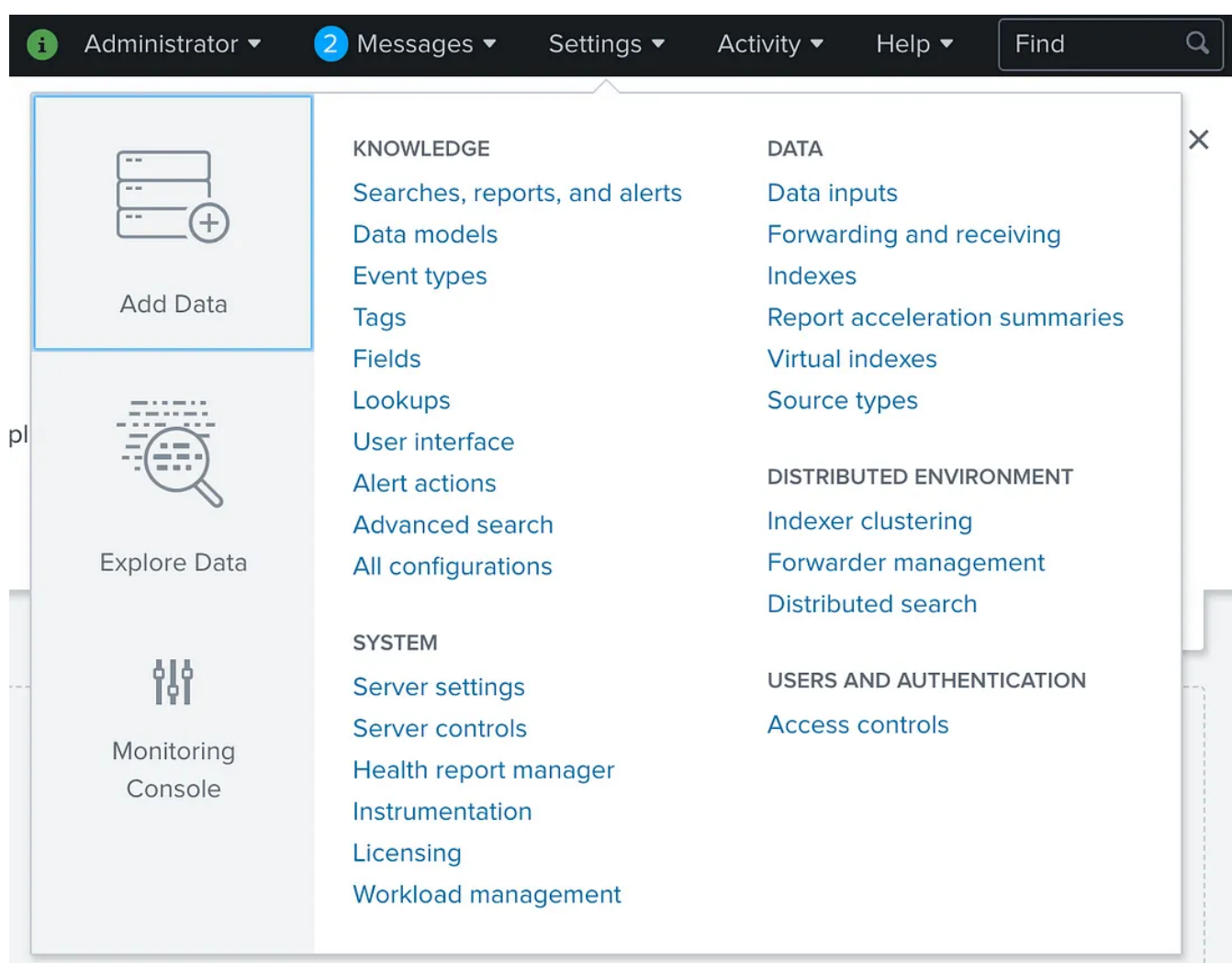
Restart Now

The result is this. The errors that exist are due to the fact that we have not set up the peer nodes.

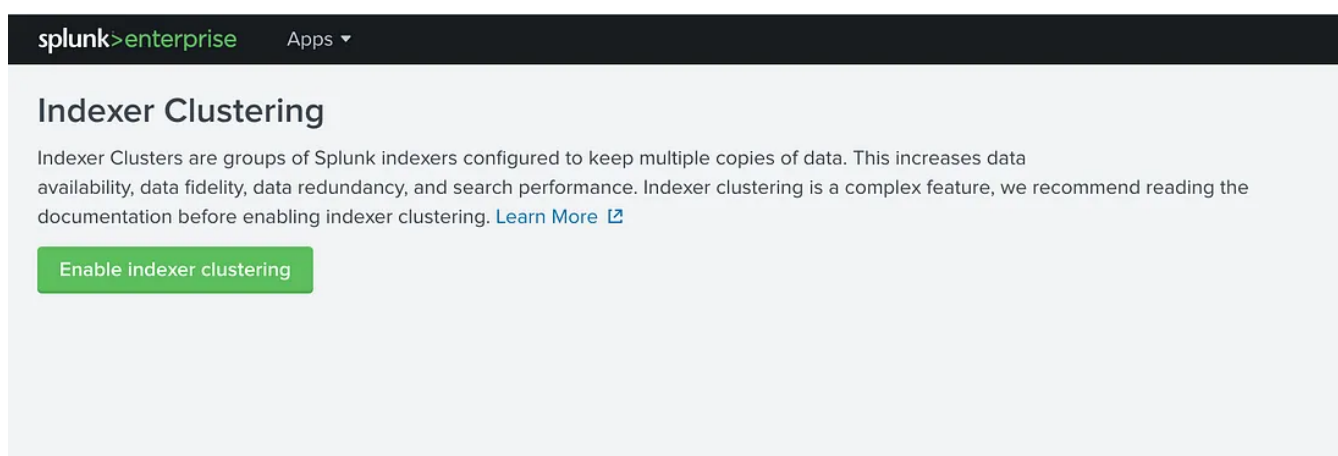
Clustering: Search Head						+ Add Cluster to be Searched Edit More Info Documentation	
Clusters searched							
Cluster Master	Searchable	Search Factor	Replication Factor	Status	Actions		
▶ https://192.168.100.23:8089	▲ Some Data is Not Searchable	▲ Not Met	▲ Not Met	✓ Up	Edit Configuration Remove Cluster		

Configuring the peer node

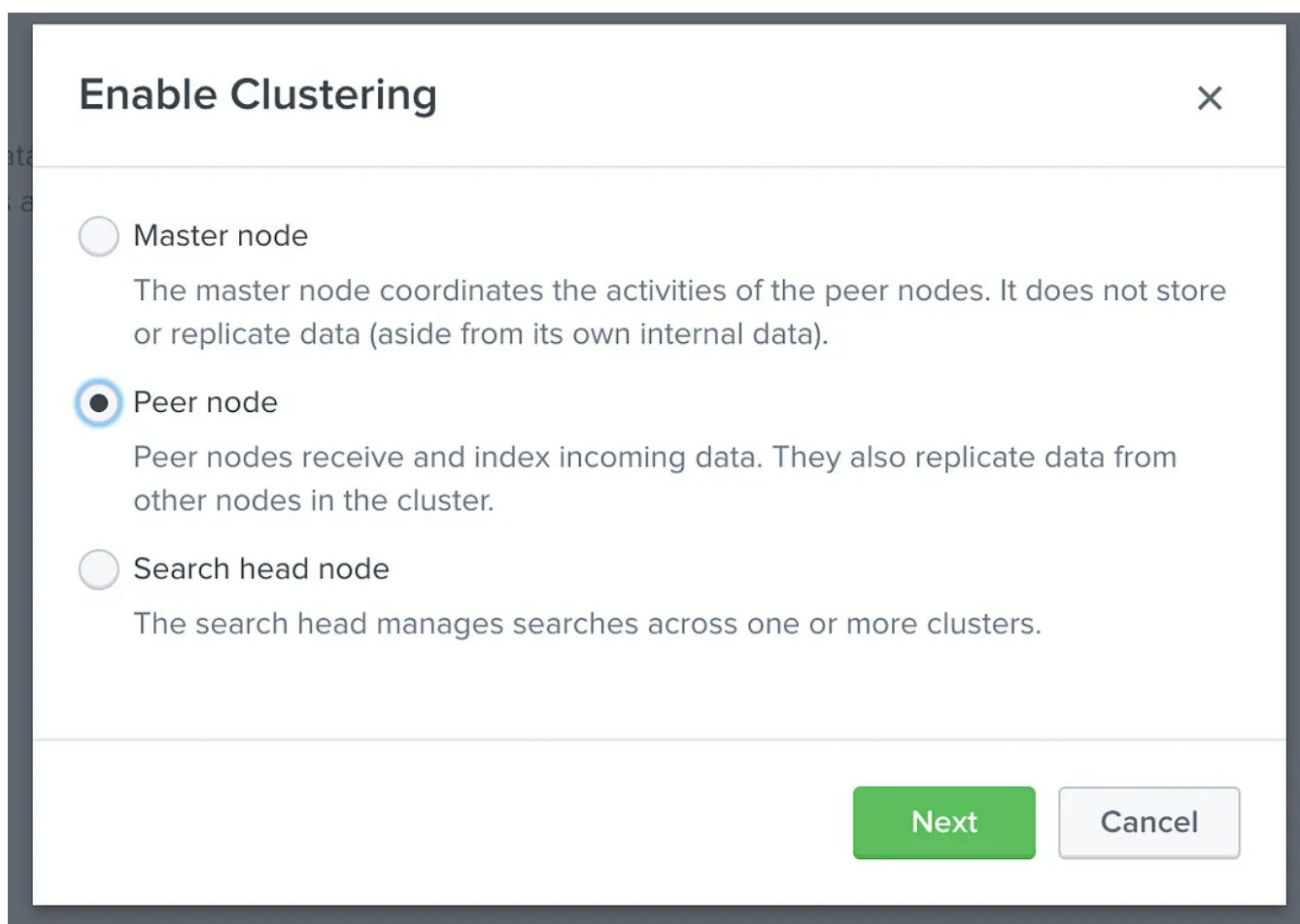
Log into the master node from the browser using the *admin* as the username and the password that was set during container creation. Once inside head to the top-right corner of the window and click *Settings > Indexer clustering*.



In the new page click *Enable Indexer clustering*



In the modal box that appears, select *peer node* and click *next*.



Under configuration, add the *Master URI*, set the replication port to *8080* and enter the security key then click enter.

Peer node configuration

Master URI

E.g. https://10.152.31.202:8089

Peer replication port

The port peer nodes use to stream data to each other (Eg: 8080).

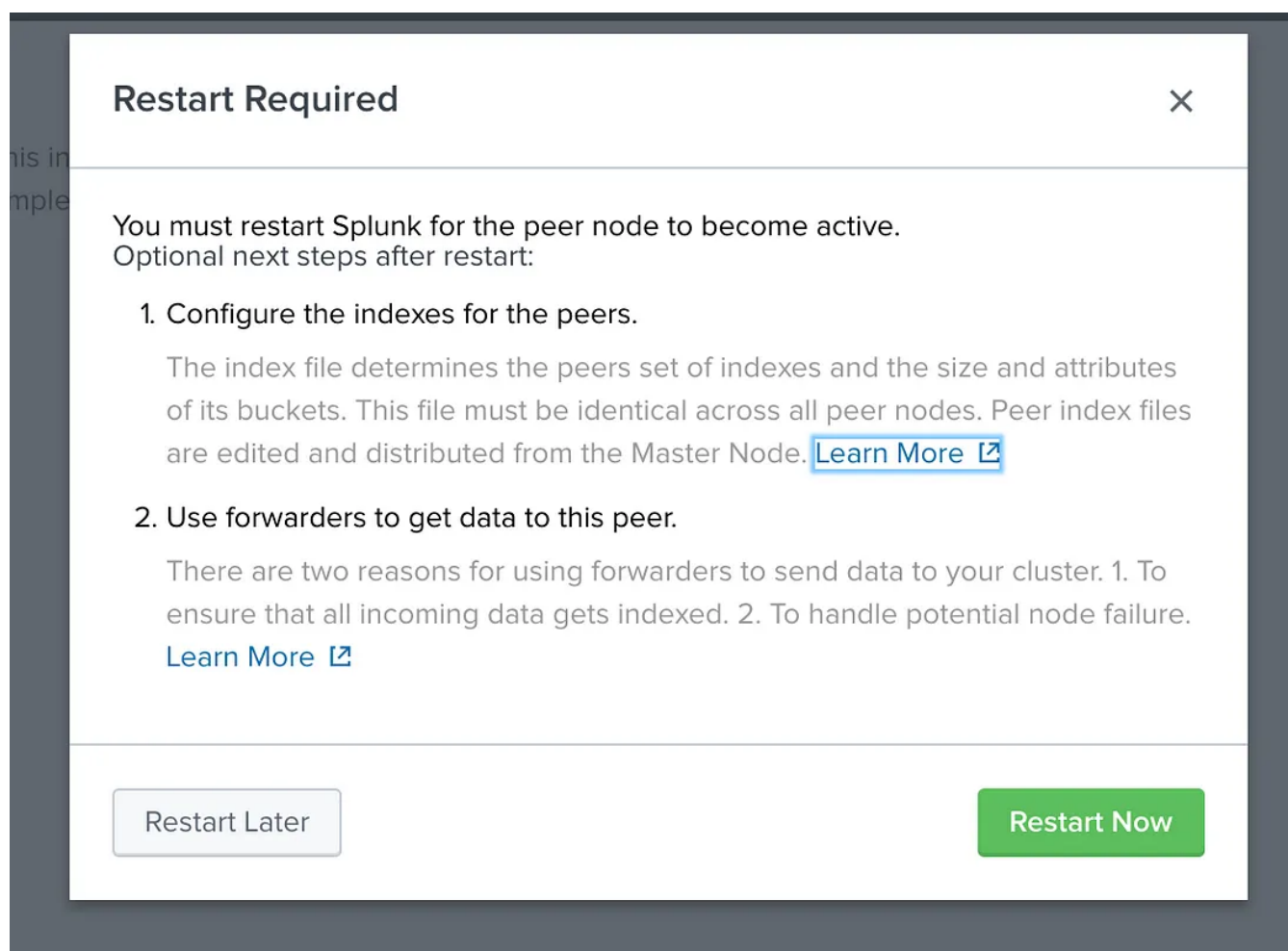
Security key

This key authenticates communication between the master and the peers and search heads.

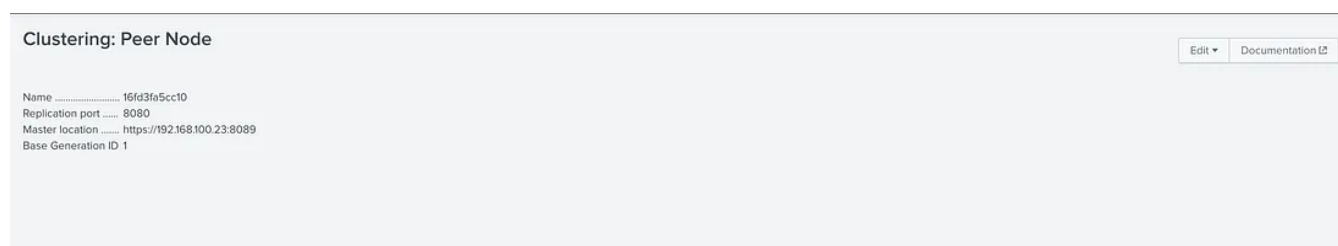
Back

Enable peer node

Restart Splunk when prompted



The result is this:



With this, the clustering is complete.

Indexes.

Data channeled to Splunk is stored in indexes and by default data channeled to Splunk is stored in the *Main* index. There are two types of indexes.

1. **Events indexes** — Events indexes impose minimal structure and can

accommodate any type of data, including metrics data. Events indexes are the default index type.

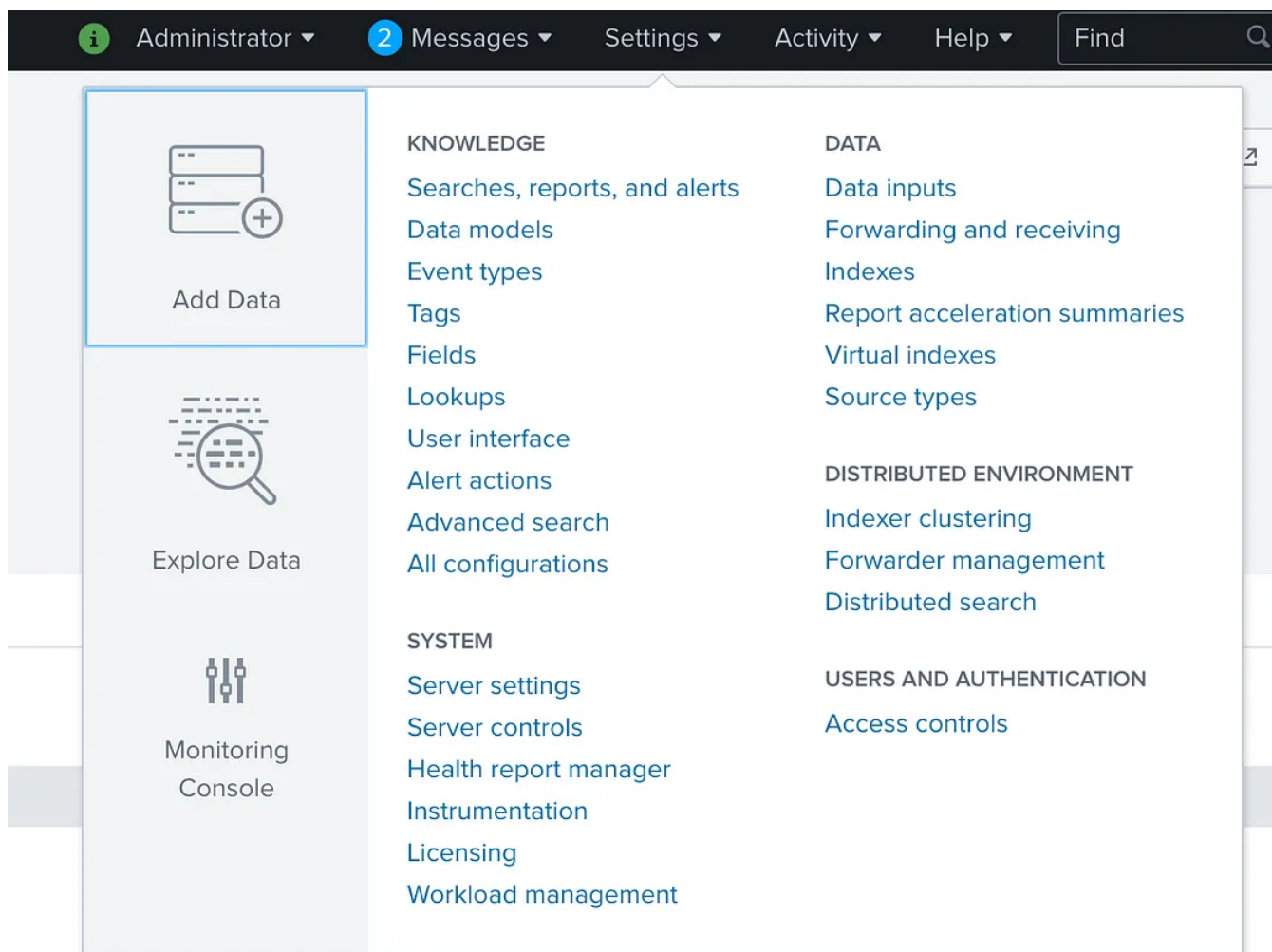
2. **Metrics indexes** — Metrics indexes use a highly structured format to handle the higher volume and lower latency demands associated with metrics data.

It is important to create custom indexes as they have the following advantages over using the main index. The advantages Include:

1. **Access management** — Access can be granted to different groups based on their roles
2. **Data retention period** — Depending on the need to store the data one can define how long data should be retained.
3. **Storage based on access frequency** — Indexes roll over from Hot/Warm(fast/frequent) to Cold buckets based on retention period/Size.
4. **Index Compression** — Data being indexed can be compressed for Storage Optimisation
5. **Faster Search** — Index defined in Search SPL will filter out events for specific index.
6. **Summary Index** — A summary index can store statistical summaries from actual index for faster search.

Creating an Index

Log into the peer node and on the top-right corner of the webpage click *settings>indexes*



In the new webpage click *New index* on the top-right corner

Indexes

A repository for data in Splunk Enterprise. Indexes reside in flat files on the Splunk Enterprise instance known as the indexer. [Learn more](#)

9 Indexes 20 per page

Name	Actions	Type	App	Current Size	Max Size	Event Count	Earliest Event	Latest Event	Home Path	Frozen Path	Status
__audit	Edit Delete Disable	Events	system	2 MB	488.28 GB	15.1K	an hour ago	a few seconds ago	\$SPLUNK_DB/audit/db	N/A	✓ Enabled
__internal	Edit Delete Disable	Events	system	5 MB	488.28 GB	30.1K	2 hours ago	a few seconds ago	\$SPLUNK_DB/_internaldb/db	N/A	✓ Enabled
__introspection	Edit Delete Disable	Events	system	6 MB	488.28 GB	3.83K	2 hours ago	a few seconds ago	\$SPLUNK_DB/_introspection/db	N/A	✓ Enabled
__telemetry	Edit Delete Disable	Events	system	1 MB	488.28 GB	0			\$SPLUNK_DB/_telemetry/db	N/A	✓ Enabled
__thefishbucket	Edit Delete Disable	Events	system	1 MB	488.28 GB	0			\$SPLUNK_DB/fishbucket/db	N/A	✓ Enabled
history	Edit Delete Disable	Events	system	1 MB	488.28 GB	0			\$SPLUNK_DB/historydb/db	N/A	✓ Enabled
main	Edit Delete Disable	Events	system	1 MB	488.28 GB	0			\$SPLUNK_DB/defaultdb/db	N/A	✓ Enabled
splunklogger	Edit Delete Enable	Events	system	0 B	488.28 GB	0			\$SPLUNK_DB/splunklogger/db	N/A	✗ Disabled
summary	Edit Delete Disable	Events	system	1 MB	488.28 GB	0			\$SPLUNK_DB/summarydb/db	N/A	✓ Enabled

In the web modal that pops up, add the name of the index, use the default setting

and click *save*. A new Index has been created.

New Index ×

General Settings

Index Name

tutorial_index

Set index name (e.g., INDEX_NAME). Search using index=INDEX_NAME.

Index Data Type

Events

Metrics

The type of data to store (event-based or metrics).

Home Path

optional

Hot/warm db path. Leave blank for default (\$SPLUNK_DB/INDEX_NAME/db).

Cold Path

optional

Cold db path. Leave blank for default (\$SPLUNK_DB/INDEX_NAME/colddb).

Thawed Path

optional

Thawed/resurrected db path. Leave blank for default (\$SPLUNK_DB/INDEX_NAME/thaweddb).

Data Integrity Check

Enable

Disable

Enable this if you want Splunk to compute hashes on every slice of your data for the purpose of data integrity.

Max Size of Entire Index

500

GB ▼

Maximum target size of entire index.

Max Size of Hot/Warm/Cold Bucket

auto

GB ▼

Maximum target size of buckets. Enter 'auto_high_volume' for high-volume indexes.

Frozen Path

optional

Frozen bucket archive path. Set this if you want Splunk to automatically archive frozen buckets.

App

_cluster ▼

Storage Optimization

Cancel

Save

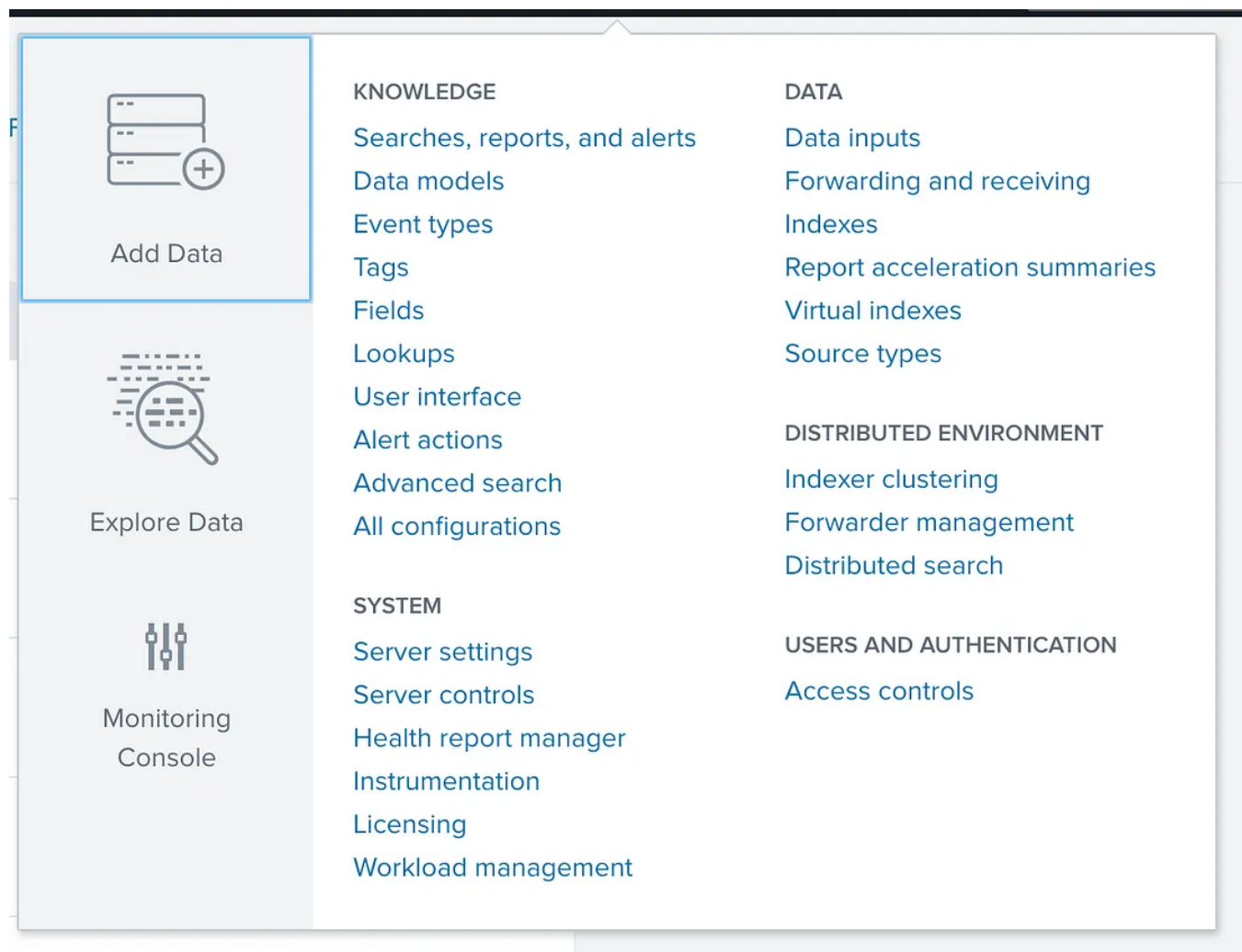
Sending Docker logs to Splunk.

Docker gives one an option of using Splunk as the logging driver that sends Docker logs to a Splunk HTTP Event Collector. This setting can be configured in two ways:

1. **Configuring the Docker settings on the host** — This will make Splunk the default logging driver for all the containers running on the host machine
2. **Specifying Splunk as the logging driver during Docker container/ docker service creation.** This sets Splunk as the logging driver for the container(s) created or to be created when scaling the docker swarm.

To achieve this, we shall create a HTTP Event Collector on the the peer node and then run a Docker container that will use our Splunk installation as the logging driver.

Log into the peer node and on the top-right corner, click settings > Data Inputs.



In the new window under HTTP Event collector click add new.

Local inputs

Type	Inputs	Actions
Files & Directories Index a local file or monitor an entire directory.	6	+ Add new
HTTP Event Collector Receive data over HTTP or HTTPS.	0	+ Add new
TCP Listen on a TCP port for incoming data, e.g. syslog.	0	+ Add new
UDP Listen on a UDP port for incoming data, e.g. syslog.	0	+ Add new
Scripts Run custom scripts to collect or generate more data.	5	+ Add new

Add a name of the HTTP event collector and click *next*.

Add Data

Select Source

Input Settings

Review

Done

< Back

Next >

[Files & Directories](#)
Upload a file, index a local file, or monitor an entire directory.

[HTTP Event Collector](#)
Configure tokens that clients can use to send data over HTTP or HTTPS.

[TCP / UDP](#)
Configure the Splunk platform to listen on a network port.

[Scripts](#)
Get data from any API, service, or database with a script.

Configure a new token for receiving data over HTTP. [Learn More](#)

Name

tutorial_http_event_collector

Source name override ?

optional

Description ?

optional

Output Group (optional)

None ▼

Enable indexer acknowledgement

☐

FAQ

> What is the HTTP Event Collector?

> How do I set up the HTTP Event Collector?

> How do I view and configure the tokens that I can use to send data to the HTTP Event Collector?

> What clients can send data to the HTTP Event Collector?

> What port and protocol does the HTTP Event Collector receive data on and how can I change that?

> What is an output group?

Under *Index*, select *tutorial_index*, click *next*

Add Data

Select Source

Input Settings

Review

Done

< Back

Review >

Input Settings

Optionally set additional input parameters for this data input as follows:

Source type

The source type is one of the default fields that the Splunk platform assigns to all incoming data. It tells the Splunk platform what kind of data you've got, so that the Splunk platform can format the data intelligently during indexing. And it's a way to categorize your data, so that you can search it easily.

Automatic

Select

New

App context

Application contexts are folders within a Splunk platform instance that contain configurations for a specific use case or domain of data. App contexts improve manageability of input and source type definitions. The Splunk platform loads all app contexts based on precedence rules. [Learn More](#)

App Context

Search & Reporting (search) ▼

Index

The Splunk platform stores incoming data as events in the selected index. Consider using a "sandbox" index as a destination if you have problems determining a source type for your data. A sandbox index lets you troubleshoot your configuration without impacting production indexes. You can always change this setting later. [Learn More](#)

Select Allowed Indexes

Available item(s)

main

summary

tutorial_index

tutorial_index

add all »

Selected item(s)« remove all

@tutorial_index

Select indexes that clients will be able to select from.

Default Index

@tutorial_index ▼

[Create a new index](#)

Review to ensure that all the settings are okay and finally click submit. The result is this page that contains a token.

✓ Token has been created successfully.
Configure your inputs by going to [Settings > Data Inputs](#)

Token Value

[Start Searching](#) Search your data now or see [examples and tutorials](#). [🔗](#)

[Add More Data](#) Add more data inputs now or see [examples and tutorials](#). [🔗](#)

[Download Apps](#) Apps help you do more with your data. [Learn more](#). [🔗](#)

[Build Dashboards](#) Visualize your searches. [Learn more](#). [🔗](#)

Even though this has been created, the HTTP event collector has not been enabled yet. To enable it goto HTTP Event collector, click global settings and enable it.

Edit Global Settings ✕

All Tokens ☒ Enabled ☐ Disabled

Default Source Type

Default Index

Default Output Group

Use Deployment Server ☐

Enable SSL ☒

HTTP Port Number [?]

[Cancel](#) [Save](#)

SSH into the 4th VM that is to be used as the APP server and run the following command:

```
docker run -p 80:80 \  
- log-driver=splunk \  
- log-opt splunk-token=f7085058-7114-4e93-9924-4653cad81dff \  
- log-opt splunk-url=https://192.168.100.23:8088 \  
- log-opt splunk-insecureskipverify=true \  
nginx
```

This starts an Nginx container that listens on port 80 and uses Splunk as its logging driver.

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

Head to the browser and type the URI of the app server and you should see a running Nginx service.

Now that the application is running, log into the search head to search the logs coming from the Docker container. To achieve this click on search and reporting on the left of the webpage and in the search window, type “index=tutorial_index”.

Here is the result.

New Search Save As Close

index="tutorial_index" Last 24 hours Q

✓ 3 events (26/11/2018 19:00:00.000 to 27/11/2018 19:50:10.000) No Event Sampling

Events (3) Patterns Statistics Visualization

Format Timeline Zoom Out Zoom to Selection Deselect 1 hour per column

List Format 20 Per Page

< Hide Fields	All Fields	i	Time	Event
SELECTED FIELDS # host 1 # source 3 # sourcetype 1 INTERESTING FIELDS # index 1 # line 3 # linecount 1 # punet 3 # splunk_server 1 # tag 1 + Extract New Fields			27/11/2018 19:49:27536	<pre>[{"_line": "2018/11/27 19:49:27 [error] 787: *1 open() \"/usr/share/nginx/html/favicon.ico" failed (2: No such file or directory), client: 172.17.0.1, server: localhost, request: \"GET /favicon.ico HTTP/1.1\", host: \"localhost:83\", referer: \"http://localhost:83/\"\", source: stderr, tag: 883d2f9f3b97\"}]\n\nShow as raw text\nhost = linuxkit-025000000001 source = http:tutorial_http_event_collector source = stderr sourcetype = httpevent</pre>
			27/11/2018 19:49:27536	<pre>[{"_line": "172.17.0.1 - - [27/Nov/2018:19:49:27 +0000] \"GET /favicon.ico HTTP/1.1\" 404 555 \"http://localhost:83/\" \"Mozilla/5.0 (Macintosh; Intel Mac OS X 10_14_0) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/70.0.3538.102 Safari/537.36\" \"-\"\", source: stdout, tag: 883d2f9f3b97\"}]\n\nShow as raw text\nhost = linuxkit-025000000001 source = http:tutorial_http_event_collector source = stdout sourcetype = httpevent</pre>
			27/11/2018 19:49:27044	<pre>[{"_line": "172.17.0.1 - - [27/Nov/2018:19:49:27 +0000] \"GET / HTTP/1.1\" 200 612 \"-\" \"Mozilla/5.0 (Macintosh; Intel Mac OS X 10_14_0) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/70.0.3538.102 Safari/537.36\" \"-\"\", source: stdout, tag: 883d2f9f3b97\"}]\n\nShow as raw text\nhost = linuxkit-025000000001 source = http:tutorial_http_event_collector source = stdout sourcetype = httpevent</pre>

From here proceed to expand the cluster by adding more peers to the cluster. You can also now use the data from your app logs to build charts, alerts and so on.

Happy Splunking

References

<https://docs.splunk.com/Documentation/Splunk/7.2.0/Indexer/Aboutclusters>

<http://docs.splunk.com/Documentation/Splunk/7.2.1/Installation/InstallonLinux>

<https://docs.splunk.com/Documentation/Splunk/7.2.1/InheritedDeployment/Ports>

<https://docs.splunk.com/Documentation/VMW/3.4.3/Installation/Networksettings>

<http://docs.splunk.com/Documentation/Splunk/7.2.1/Indexer/Aboutmanagingindexes>

<https://answers.splunk.com/answers/614092/what-is-the-meaning-of-index.html>

<https://docs.docker.com/config/containers/logging/splunk/>

Docker

Splunk

Logging

Cluster

Logs

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[about 6 years ago](#)



[Thank you for such a good write up!](#)
[Few suggestions:](#)

1



[Balakrishna Sriramineni](#)
[almost 6 years ago](#)



Hi,

One suggestion:

We shouldn't directly create an index on the indexers. I do understand that you have only one indexer connected to Cluster master and RF and SF is set to "1". But this would be misleading the users when they try

to set up an IDXC.....

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Introduction:

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```
2  when
3    NOT (contains(to_string($message.message), "validation result")
4  then
5    let val = grok(
6      pattern:"%{NOTSPACE:host} %{NOTSPACE:logSystem} %{SYSLOGTIMES
7      value:to_string($message.message),
8      only_named_captures: true);
9
10   set_field(
11     field:"host".
12
```

Rule source, see quick reference for more information.

Rule Simulation (Optional)

JSON

Key Value

Simple Message

simple message field



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